

JESS Thermodynamic Database v8.9

Niobium

24-Jan-23

Reaction No. 4804							
$\text{NbO}+3 + 2\text{H}^+ + 2\text{e}^- = \text{Nb}+3 + \text{H}_2\text{O}$							
Note: NbO+3 unhydrated is inappropriate							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-67.7(3SF)	0	EES	
2	18	0	HCl	E:-0.3(0.05SD)	0	MEF	775

Reaction No. 4805							
$\text{Nb}+3 + 3\text{e}^- = \text{Nb(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	lgK:-56.(2SF)	3	ENS	140
2	25	0	Inf. Dilution	E:-1.099(4SF)	5	CRV	6330
3	25	0	Inf. Dilution	E:-1.1(2SF)	4	CRV	22551

Reaction No. 13718							
$\text{Nb}+5_{\text{OH}-1(4)} + \text{H}^+(3)_{\text{34DHB}-3} = \text{Nb}+5_{\text{H}^+(2)_{\text{OH}-1(4)}_{\text{34DHB}-3}} + \text{H}^+$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22	0.1	Unknown	lgK:2.3(2SF)	4	MSP	931
Note (1): NbO2+1 notation in [931]							

Reaction No. 13719							
$\text{Nb}+5_{\text{H}^+(2)_{\text{OH}-1(4)}_{\text{34DHB}-3}} + \text{H}^+(3)_{\text{34DHB}-3} = \text{Nb}+5_{\text{H}^+(4)_{\text{OH}-1(4)}_{\text{34DHB}-3(2)}} + \text{H}^+$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22	0.1	Unknown	lgK:1.3(2SF)	4	MSP	931
Note (1): NbO2+1 notation in [931]							

Reaction No. 13720							
$\text{Nb}+5_{\text{OH}-1(4)} + 3\text{H}^+_{\text{34DHB}-3} + 4\text{H}^+ = \text{Nb}+5_{\text{H}^+(7)_{\text{OH}-1(4)}_{\text{34DHB}-3(3)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:22	0.1	Unknown	lgK:63.1(3SF)	4	MSP	931
Note (1): NbO2+1 notation in [931]							

Reaction No. 40390							
$\text{Nb+5_OH-1(4)} + \text{H+1(2)_Oxalic-2} = \text{Nb+5_H+1_OH-1(4)_Oxalic-2} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4.5	(H,Na)Cl	lgK:3.55(3SF)	3	MDS	931
2	20	4.5	(H,Na)Cl	lgK:3.48(3SF)	2	MDS	30350
Note (2): Doubts about how the reaction is written; number of OH in complex in doubt							
3	20	4.5	(H,Na)NO3	lgK:3.61(3SF)	2	MDS	30350
Note (3): Doubts about how the reaction is written; number of OH in complex in doubt							
4	25	0	Inf. Dilution	lgK:3.0(2SF)	1	ESC	

Reaction No. 40391							
$\text{Nb+5_OH-1(4)} + 2\text{<H+1(2)_Oxalic-2>} = \text{Nb+5_OH-1(2)_Oxalic-2(2)} + 2\text{<H+1>} + 2\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4.5	(H,Na)Cl	lgK:5.13(3SF)	0	MDS	931
2	20	4.5	(H,Na)Cl	lgK:5.18(3SF)	2	MDS	30350
Note (2): Doubts about how the reaction is written; number of OH in complex in doubt							
3	20	4.5	(H,Na)NO3	lgK:5.09(3SF)	2	MDS	30350
Note (3): Doubts about how the reaction is written; number of OH in complex in doubt							
4	25	0	Inf. Dilution	lgK:5.7(2SF)	1	ELC	

Reaction No. 40392							
$\text{Nb+5_OH-1(4)} + 2\text{<H+1_Oxalic-2>} = \text{Nb+5_OH-1(2)_Oxalic-2(2)} + 2\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Unknown	lgK:12.11(4SF)w	0	MPH	931
Note (1): Reaction written $\text{Nb+5_OH-1(5)(s)} + 3\text{<H+1>} + 2\text{<Oxalic-2>}$ in orig. pub [30355]							

Reaction No. 40393							
$\text{Nb+5_OH-1(4)} + 2\text{<H+1_Oxalic-2>} + \text{Oxalic-2} = \text{Nb+5_OH-1(2)_Oxalic-2(3)} + 2\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Unknown	lgK:17.15(4SF)w	0	MPH	931
Note (1): Reaction written $\text{Nb+5_OH-1(5)(s)} + 3\text{<H+1>} + 3\text{<Oxalic-2>}$ in orig. pub. [30355]							

Reaction No. 42855							
$2\text{<Oxalic-2>} + \text{Nb+5_OH-1(2)} = \text{Nb+5_OH-1(2)_Oxalic-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	0.5	Unknown	lgK:35.9(3SF)	0	MSL	906

Note (1): NbO+3 in [906]							
2	25	0	Inf. Dilution	lgK:33.8(3SF)	1	ELC	

Reaction No. 43348							
Nb+5_OH-1(4) + H+1(2)_Asp-2 = Nb+5_H+1(2)_OH-1(4)_Asp-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HNO3	lgK:4.82(3SF)	4	MDS	906
Note (1): NbO2+1 notation in original 906							

Reaction No. 43703							
Ascorbic-2 + Nb+5 = Nb+5_Ascorbic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:9.4(2SF)	0	MSP	906
Note (1): Unreal species Nb+5							

Reaction No. 44435							
Nb+5_Cl-1(5)_OH-1(2) + 2<H+1(3)_SalHydroxam-2> = Nb+5_H+1(6)_Cl-1(5)_OH-1(2)_SalHydroxam-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Unknown	lgK:4.6(0.05SD)	0	MSP	906

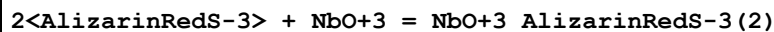
Reaction No. 44860							
EDTA-4 + Nb+5 = Nb+5_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.4	KCl	lgK:39.4(3SF)	0	MPL	906
Note (1): Unreal species Nb+5; looks like Sochevanov (1969) [30456]; look at Rxn 80182 (Volkova 30356)							

Reaction No. 44861							
Nb+5_OH-1(4) + H+1(3)_EDTA-4 = Nb+5_H+1(3)_OH-1(4)_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	KNO3	lgK:10.54(4SF)	4	MDS	906
Note (1): Rf wrong Koch H. Kupsch Z Naturforsch B 1969 24 398 NbO2+1 in [906]							
2	25	0	Inf. Dilution	lgK:10.4(3SF)	1	ESC	

Reaction No. 45059							
NbO+3_Tartaric-2 + PAR-2 = NbO+3_PAR-2_Tartaric-2							
Note: NbO+3 unhydrated is inappropriate							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:21.2 (0.05SD)	0	MSP	906

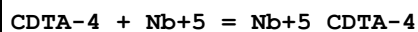
Reaction No. 45234



Note: NbO+3 unhydrated is inappropriate

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:8. (0.3SD)	0	MSP	906

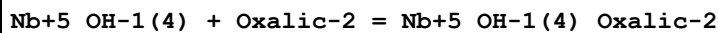
Reaction No. 45249



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	22	2	Unknown	lgK:15.60 (4SF)	0	MPL	906

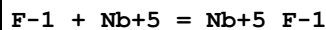
Note (1): Unreal species Nb+5

Reaction No. 45386



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	0.5	Unknown	lgK:9.08 (3SF)	4	MSL	906
2	25	0	Inf. Dilution	lgK:9.3 (2SF)	1	ESC	

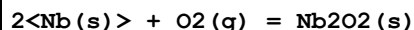
Reaction No. 53172



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	17	H2SO4	lgK:7.12 (3SF)	0	MSP	5398

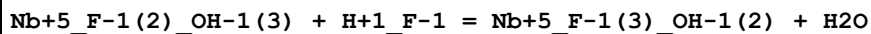
Note (1): Nb+5 unreal species

Reaction No. 53728



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-422.6 (4SF)	4	CRV	22971
2	25	0	Inf. Dilution	dH:-454.5 (4SF)	4	CRV	22971

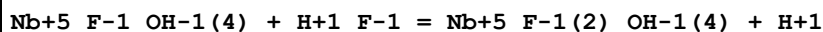
Reaction No. 54122



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18	3	HNO3	lgK:4.2 (2SF)	4	MSL	31278

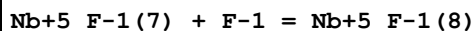
2	25	0	Inf. Dilution	lgK:4.(1SF)	1	ESC	
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Reaction No. 54123



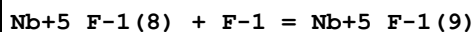
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.3(2SF)	1	ESC	
2	18	0.5	HNO3	lgR:3.6(2SF)	4	MSL	31278

Reaction No. 54480



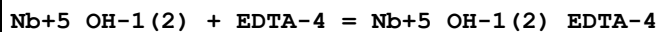
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO4	lgK:3.08(3SF)	3	MIS	437
2	25	0.5	NaClO4	lgK:3.1(2SF)	3	MIS	30149

Reaction No. 54483



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO4	lgK:4.0(2SF)	3	MIS	437
2	25	0.5	NaClO4	lgK:4.0(2SF)	3	MIS	30149

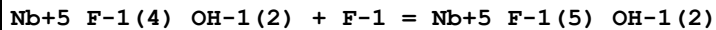
Reaction No. 60181



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	Unknown	lgK:40.78(4SF)	0	MPL	931

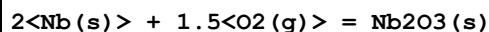
Note (1): Probably Volkova & Sochevanov (1967) [30357]; see: Nb+5_OH-1(5) + H+1(4)_EDTA-4

Reaction No. 60686



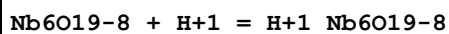
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO4	lgK:4.51(3SF)	3	MIS	5398
2	25	0.5	NaClO4	lgK:4.5(2SF)	3	MIS	30149
3	25	3	KCl	lgK:2.51(3SF)	3	MPT	5398
4	25	3	KCl	lgK:2.51(0.01SD)	5	MPT	12517
5	25	3	Unknown	lgK:2.51(3SF)	3	CRV	2556

Reaction No. 61488



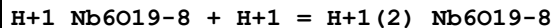
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-422.06(5SF)	3	CRV	31291

Reaction No. 65907



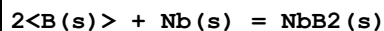
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.11(0.14SD)	3	MGL	847
Note (1): Application of Bromley theory							
2	25	0	Inf. Dilution	lgK:14.29(4SF)	4	CRV	31495
3	25	3	KCl	lgK:13.63(0.04SD)	3	MGL	847
Note (3): pH from 9.7 to 13.1							
4	25	3	KCl	lgK:13.4(0.1SD)	4	MSP	30111
Note (4): pH from 12 to 14.5							
5	25	3	KCl	lgK:13.8(0.2SD)	4	MGL	30155
Note (5): pH from 10.7 to 12.25							
6	25	3	Unknown	lgK:13.8(3SF)	4	CRV	552
7	25	3	KCl	dH:-95.(2SF) kJ	4	MTD	30111

Reaction No. 65908



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.23(4SF)	4	CRV	31495
2	25	3	KCl	lgK:10.88(0.05SD)	3	MGL	30155
Note (2): pH from 10.7 to 12.25							
3	25	3	Unknown	lgK:10.88(4SF)	3	CRV	552

Reaction No. 67063



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:43.44(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:43.17(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:32.24(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:25.68(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:21.31(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-59.26(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-60.(2SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-60.0(3SF)	4	MTD	6422

9	25	0	Inf. Dilution	dH:-61.(2SF)	4	CRV	22971
10	127	0	Inf. Dilution	dH:-60.0(3SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-60.0(3SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-60.0(3SF)	0	MTD	6422

Reaction No. 67064							
2.5<Br2(1)> + Nb(s) = Nb+5_Br-1(5)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:89.22(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:88.62(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:62.39(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:45.96(4SF)	1	ENS	6422
5	254	0	Inf. Dilution	lgK:42.60(4SF)	1	ENS	6422
6	25	0	Inf. Dilution	dG:-121.7(4SF)	3	EFE	6422
7	25	0	Inf. Dilution	dG:-122.(3SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-133.0(4SF)	3	MTD	6422
9	25	0	Inf. Dilution	dH:-132.8(4SF)	4	CRV	22971
10	127	0	Inf. Dilution	dH:-150.7(4SF)	0	MTD	6422
11	227	0	Inf. Dilution	dH:-150.0(4SF)	0	MTD	6422
12	254	0	Inf. Dilution	dH:-149.8(4SF)	0	MTD	6422

Reaction No. 67065							
C(s) + 2<Nb(s)> = Nb2C(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.53(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:32.32(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:24.06(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:19.12(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:15.84(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-44.37(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-47.(2SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-45.4(3SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-47.(2SF)	4	CRV	22971
10	127	0	Inf. Dilution	dH:-45.3(3SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-45.1(3SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-44.9(3SF)	0	MTD	6422

Reaction No. 67066							
$\text{Cl}_2(\text{g}) + \text{Nb}(\text{s}) = \text{NbCl}_2(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:63.89 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:63.45 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:45.75 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:35.18 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:28.15 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-87.16 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-97.3 (3SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-97.0 (3SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-96.6 (3SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-96.2 (3SF)	0	MTD	6422

Reaction No. 67067							
$1.165\text{Cl}_2(\text{g}) + \text{Nb}(\text{s}) = \text{NbCl}_{2.33}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:74.47 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:73.96 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:53.34 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:41.01 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:32.82 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-101.6 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-102. (3SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-113.4 (4SF)	2	MTD	6422
9	25	0	Inf. Dilution	dH:-114. (3SF)	4	CRV	22971
10	127	0	Inf. Dilution	dH:-113.0 (4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-112.6 (4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-112.1 (4SF)	0	MTD	6422

Reaction No. 67068							
$1.335\text{Cl}_2(\text{g}) + \text{Nb}(\text{s}) = \text{NbCl}_{2.67}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:83.98 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:83.40 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:60.01 (4SF)	2	ENS	6422

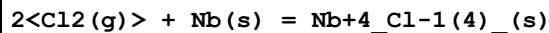
4	227	0	Inf. Dilution	lgK:46.02 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:36.73 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-114.6 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-115. (3SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-128.6 (4SF)	2	MTD	6422
9	25	0	Inf. Dilution	dH:-129. (3SF)	4	CRV	22971
10	127	0	Inf. Dilution	dH:-128.2 (4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-127.8 (4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-127.3 (4SF)	0	MTD	6422

Reaction No. 67069							
1.5<Cl ₂ (g)> + Nb(s) = Nb+3_Cl-1(3)_ (s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:90.20 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:89.57 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:64.29 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:49.18 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:39.14 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-123.05 (5SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-124. (3SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-139.0 (4SF)	2	MTD	6422
9	25	0	Inf. Dilution	dH:-140. (3SF)	4	CRV	22971
10	127	0	Inf. Dilution	dH:-138.5 (4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-138.0 (4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-137.5 (4SF)	0	MTD	6422

Reaction No. 67070							
1.565<Cl ₂ (g)> + Nb(s) = NbCl ₃ .13(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:93.84 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:93.18 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:66.87 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:51.14 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:40.70 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-128.0 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-127. (3SF)	4	CRV	22971

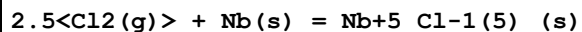
8	25	0	Inf. Dilution	dH:-144.7 (4SF)	2	MTD	6422
9	25	0	Inf. Dilution	dH:-144. (3SF)	4	CRV	22971
10	127	0	Inf. Dilution	dH:-144.2 (4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-143.6 (4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-143.1 (4SF)	0	MTD	6422

Reaction No. 67071



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:106.1 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:105.3 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:75.16 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:57.14 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:45.17 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-144.7 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-145. (3SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-166.0 (4SF)	2	MTD	6422
9	25	0	Inf. Dilution	dH:-166.0 (4SF)	4	CRV	22971
10	127	0	Inf. Dilution	dH:-165.3 (4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-164.6 (4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-163.8 (4SF)	0	MTD	6422

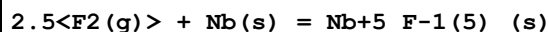
Reaction No. 67072



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:119.9 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:119.0 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:84.36 (4SF)	4	ENS	6422
4	205	0	Inf. Dilution	lgK:67.31 (4SF)	2	ENS	6422
5	25	0	Inf. Dilution	dG:-683.2 (4SF) kJ	3	CRV	6330
6	25	0	Inf. Dilution	dG:-163.5 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-163.2 (4SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-797.5 (4SF) kJ	3	CRV	6330
9	25	0	Inf. Dilution	dH:-190.6 (4SF)	4	MTD	6422
10	25	0	Inf. Dilution	dH:-190.5 (4SF)	4	CRV	22971
11	127	0	Inf. Dilution	dH:-189.7 (4SF)	4	MTD	6422

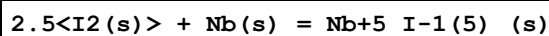
12	205	0	Inf. Dilution	dH:-189.1(4SF)	2	MTD	6422
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Reaction No. 67073



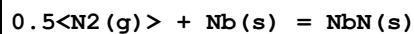
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:297.8(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:295.8(4SF)	0	ENS	6422
3	77.7	0	Inf. Dilution	lgK:250.1(4SF)	4	ENS	6422
4	25	0	Inf. Dilution	dG:-406.2(4SF)	4	EFE	6422
5	25	0	Inf. Dilution	dG:-406.1(4SF)	4	CRV	22971
6	25	0	Inf. Dilution	dH:-433.5(4SF)	4	MTD	6422
7	25	0	Inf. Dilution	dH:-433.5(4SF)	4	CRV	22971
8	77.7	0	Inf. Dilution	dH:-434.4(4SF)	4	MTD	6422

Reaction No. 67074



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:47.91(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:47.62(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:35.74(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:26.60(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:19.41(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-65.36(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-64.2(3SF)	4	MTD	6422
8	25	0	Inf. Dilution	dH:-64.6(3SF)	4	CRV	22971
9	127	0	Inf. Dilution	dH:-73.9(3SF)	4	MTD	6422
10	227	0	Inf. Dilution	dH:-99.3(3SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-97.9(3SF)	0	MTD	6422

Reaction No. 67075



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:36.09(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:35.83(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:25.60(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:19.46(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:15.37(4SF)	0	ENS	6422

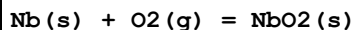
6	25	0	Inf. Dilution	dG:-49.23 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-49.7 (3SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-56.2 (3SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-56.4 (3SF)	4	CRV	22971
10	127	0	Inf. Dilution	dH:-56.2 (3SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-56.1 (3SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-56.0 (3SF)	0	MTD	6422

Reaction No. 67076							
0.5<N2(g)> + 2<Nb(s)> = Nb2N(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.59 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:38.32 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:27.42 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:20.89 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:16.55 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-52.65 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-59.9 (3SF)	4	MTD	6422
8	25	0	Inf. Dilution	dH:-60.4 (3SF)	4	CRV	22971
9	127	0	Inf. Dilution	dH:-59.8 (3SF)	4	MTD	6422
10	227	0	Inf. Dilution	dH:-59.7 (3SF)	4	MTD	6422
11	327	0	Inf. Dilution	dH:-59.6 (3SF)	0	MTD	6422

Reaction No. 67077							
Nb(s) + 0.5<O2(g)> = NbO(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:68.67 (4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:68.21 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:49.95 (4SF)	1	ENS	6422
4	227	0	Inf. Dilution	lgK:39.00 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:31.71 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-93.68 (4SF)	0	EFE	6422
7	25	0	Inf. Dilution	dG:-91. (2SF)	4	CRV	22971
8	25	0	Inf. Dilution	dG:-90.529 (5SF)	3	CRV	31291
9	25	0	Inf. Dilution	dH:-100.3 (4SF)	2	MTD	6422
10	25	0	Inf. Dilution	dH:-98. (2SF)	2	CRV	22971

11	127	0	Inf. Dilution	dH:-100.2(4SF)	4	MTD	6422
12	227	0	Inf. Dilution	dH:-100.1(4SF)	4	MTD	6422
13	327	0	Inf. Dilution	dH:-100.0(4SF)	0	MTD	6422

Reaction No. 67078



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:129.5(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:128.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:94.05(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:73.32(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:59.51(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-176.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-177.1(4SF)	4	CRV	22971
8	25	0	Inf. Dilution	dG:-176.98(5SF)	3	CRV	31291
9	25	0	Inf. Dilution	dG:-740.5(0.100SD)kJ	4	CRV	32222
10	25	0	Inf. Dilution	dH:-190.0(4SF)	4	MTD	6422
11	25	0	Inf. Dilution	dH:-190.4(4SF)	4	CRV	22971
12	127	0	Inf. Dilution	dH:-189.8(4SF)	1	MTD	6422
13	227	0	Inf. Dilution	dH:-189.6(4SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-189.3(4SF)	0	MTD	6422

Reaction No. 67079



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:309.4(4SF)	0	ENS	6422
2	27	0	Inf. Dilution	lgK:307.3(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:224.7(4SF)	0	ENS	6422
4	227	0	Inf. Dilution	lgK:175.1(4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:142.1(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-422.1(4SF)	0	EFE	6422
7	25	0	Inf. Dilution	dG:-424.6(4SF)	0	CRV	28287

Note (7): average of four values

8	25	0	Inf. Dilution	dG:-422.9(0.7SD)	2	EFE	30139
9	25	0	Inf. Dilution	dG:-1764.0(1.9SD)kJ	2	EOR	30153

Note (9): Original Russian reference given

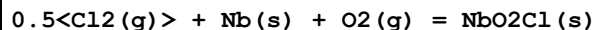
10	25	0	Inf. Dilution	dG:-1766.0(0.100SD)kJ	4	CRV	32222
11	25	0	Inf. Dilution	dH:-454.0(4SF)	0	MTD	6422
12	25	0	Inf. Dilution	dH:-455.2(0.6SD)	4	MCL	30139
13	25	0	Inf. Dilution	dH:-1897.4(1.7SD)kJ	3	EOR	30153
Note (13): Original Russian reference given							
14	127	0	Inf. Dilution	dH:-453.7(4SF)	0	MTD	6422
15	227	0	Inf. Dilution	dH:-453.1(4SF)	0	MTD	6422
16	327	0	Inf. Dilution	dH:-452.5(4SF)	0	MTD	6422

Reaction No. 67080							
$\text{Cl}_2(\text{g}) + \text{Nb}(\text{s}) + 0.5\text{O}_2(\text{g}) = \text{NbOCl}_2(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:123.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:122.3(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:88.61(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:68.46(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:55.08(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-168.0(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-185.1(4SF)	2	MTD	6422
8	25	0	Inf. Dilution	dH:-186.(3SF)	4	CRV	22971
9	127	0	Inf. Dilution	dH:-184.6(4SF)	1	MTD	6422
10	227	0	Inf. Dilution	dH:-184.0(4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-183.4(4SF)	0	MTD	6422

Reaction No. 67081							
$1.5\text{Cl}_2(\text{g}) + \text{Nb}(\text{s}) + 0.5\text{O}_2(\text{g}) = \text{NbOCl}_3(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:136.8(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:135.8(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:97.59(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:74.74(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:59.57(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-186.6(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-189.(3SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-210.2(4SF)	2	MTD	6422
9	25	0	Inf. Dilution	dH:-211.(3SF)	4	CRV	22971

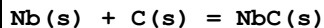
10	127	0	Inf. Dilution	dH:-209.4 (4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-208.7 (4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-207.9 (4SF)	0	MTD	6422

Reaction No. 67082



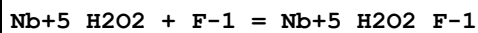
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:158.4 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:157.4 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:114.6 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:89.05 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:72.04 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-216.2 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-235.0 (4SF)	2	MTD	6422
8	127	0	Inf. Dilution	dH:-234.4 (4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-233.8 (4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-233.2 (4SF)	0	MTD	6422

Reaction No. 67244



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.98 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:23.83 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:17.80 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:14.19 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:11.79 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-32.72 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-33. (2SF)	4	CRV	22971
8	25	0	Inf. Dilution	dH:-33.2 (3SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-33. (2SF)	4	CRV	22971
10	127	0	Inf. Dilution	dH:-33.1 (3SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-33.0 (3SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-32.9 (3SF)	0	MTD	6422

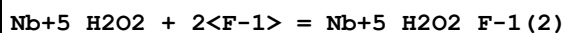
Reaction No. 68860



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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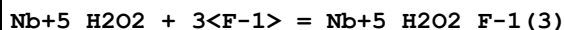
1	23		H2SO4(1):Water 20:80Vol	lgK:2.62 (3SF)	0	MSP	6680
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Reaction No. 68861



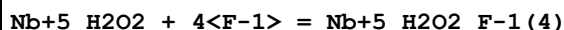
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		H2SO4(1):Water 20:80Vol	lgK:1.71 (3SF)	0	MSP	6680

Reaction No. 68862



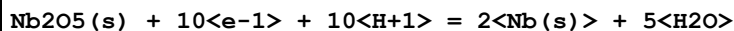
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		H2SO4(1):Water 20:80Vol	lgK:1.58 (3SF)	0	MSP	6680

Reaction No. 68863



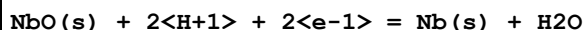
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23		H2SO4(1):Water 20:80Vol	lgK:1.51 (3SF)	0	MSP	6680

Reaction No. 71070



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-102. (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.644 (3SF)	0	CRV	6330
3	25	0	Inf. Dilution	E:-0.65 (2SF)	0	CRV	22551

Reaction No. 71071



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-25.1 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.733 (3SF)	0	CRV	6330
3	25	0	Inf. Dilution	E:-0.733 (3SF)	0	CRV	28287

Reaction No. 71072



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-46.4 (3SF)	1	ELC	

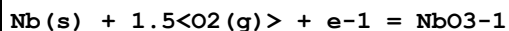
2	25	0	Inf. Dilution	E:-0.690 (3SF)	0	CRV	6330
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Reaction No. 71073



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-21.4 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.646 (3SF)	0	CRV	6330

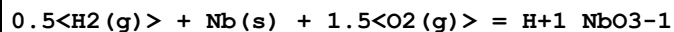
Reaction No. 72392



Note: NbO3-1 unhydrated is inappropriate

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-227.100 (6SF)	4	CRV	9029
2	25	1	Unknown	dG:-932.100 (0.100SD) kJ	4	CRV	32222
3	25	0	Inf. Dilution	dH:-255.300 (6SF)	4	CRV	9029

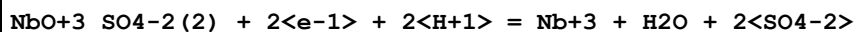
Reaction No. 72393



Note: NbO3-1 unhydrated is inappropriate

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-236.800 (6SF)	4	CRV	9029
2	25	0	Inf. Dilution	dH:-257.950 (6SF)	4	CRV	9029

Reaction No. 74262

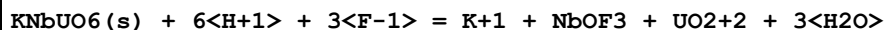


Note: NbO+3 unhydrated is inappropriate

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.1 (1SF)	0	CRV	22551

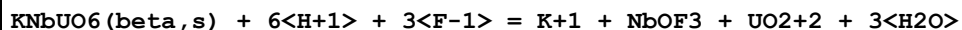
Note (1): Approximate value in source

Reaction No. 75051



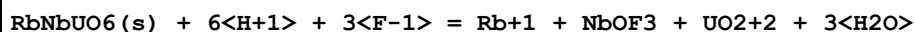
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:265.5 (4SF) kJ	1	MCL	22973

Reaction No. 75052



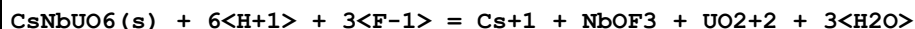
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:215.7(4SF)kJ	1	MCL	22973

Reaction No. 75053



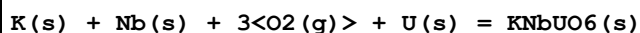
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:229.8(4SF)kJ	1	MCL	22973

Reaction No. 75054



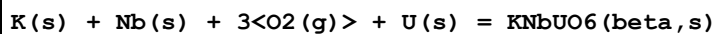
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:250.1(4SF)kJ	1	MCL	22973

Reaction No. 75059



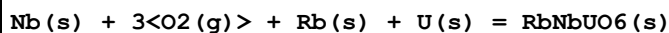
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2590.(4SF)kJ	1	MCL	22973

Reaction No. 75060



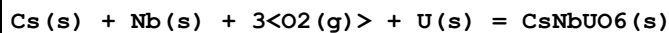
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2619.(4SF)kJ	1	MCL	22973

Reaction No. 75061



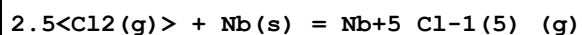
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2633.(4SF)kJ	1	MCL	22973

Reaction No. 75062



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2638.(4SF)kJ	1	MCL	22973

Reaction No. 75629



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-646.0(4SF)kJ	3	CRV	6330
2	25	0	Inf. Dilution	dH:-703.7(4SF)kJ	3	CRV	6330

Reaction No. 77487							
$\text{Nb(s)} + \text{O}_2(\text{g}) - e^- = \text{NbO}_2 + 1$							
Note: NbO ₂ +1 unhydrated is inappropriate							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-163.0 (4SF)	3	CRV	28287

Reaction No. 77488							
$\text{Nb(s)} + 0.5\text{O}_2(\text{g}) - 2e^- = \text{NbO} + 2$							
Note: NbO+2 unhydrated is inappropriate							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-101.7 (4SF)	3	CRV	28287

Reaction No. 77491							
$\text{Nb} + 3e^- + \text{H}_2\text{O} = \text{NbO(s)} + 2\text{H}^+$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-30.7 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.9 (1SF)	0	CRV	28287

Reaction No. 77492							
$\text{NbO}_2(\text{s}) + e^- + 4\text{H}^+ = \text{Nb} + 3\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.4 (2SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.4 (1SF)	0	CRV	28287

Reaction No. 77493							
$\text{Nb}_2\text{O}_5(\text{s}) + 2e^- + 2\text{H}^+ = 2\text{NbO}_2(\text{s}) + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.2 (2SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.248 (3SF)	0	CRV	28287

Reaction No. 80132							
$2\text{H}_2(\text{g}) + \text{Nb(s)} + 2\text{O}_2(\text{g}) - e^- = \text{Nb}_5\text{O}_{14} + 4\text{H}^+$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:210. (3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-365.440 (6SF)	0	MDG	30153
3	25	0	Inf. Dilution	dH:-421.760 (6SF)	0	MTD	30153

Reaction No. 80141							
$2.5\text{H}_2(\text{g}) + \text{Nb}(\text{s}) + 2.5\text{O}_2(\text{g}) = \text{Nb}_5\text{OH-1}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-340.100(3SF)	3	MDG	30153
2	25	1	Unknown	dG:-1208.6(0.100SD)kJ	4	CRV	32222
3	25	0	Inf. Dilution	dH:-394.100(3SF)	3	MTD	30153

Reaction No. 80142							
$\text{Nb}_5\text{OH-1}(\text{s})_{(\text{am.}, \text{s})} = \text{Nb}_5\text{OH-1}(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:-4.8(2SF)	0	EOD	403
Note (1): Calculated from Babko solubility [63BLN_30106]							
2	25	0	Inf. Dilution	lgK:-7.9(2SF)	3	EDH	
Note (2): Value from ref [31495] selected and extrapolated of Inf. Dil.							
3	25	0	Inf. Dilution	lgK:-4.(1SF)	1	ELC	
4	25	0.1	NaCl	lgK:-8.(1SF)	0	MSL	31495
Note (4): value/2; orig. $2\text{Nb}(\text{OH})_5 = \text{Nb}_2\text{O}_5(\text{pr}) + 5\text{H}_2\text{O}$, here amorphous prec. $\text{Nb}_5\text{OH-1}(\text{s})$							
5	25	1	KNO3	lgK:-4.855(4SF)	0	EOD	847
Note (5): Calc. from Babko sol. [63BLN_30106]; $\text{Nb}_2\text{O}_5(\text{s}) + 5\text{H}_2\text{O} = 2\text{Nb}(\text{OH})_5$ lgK=-9.71, here divided by 2							

Reaction No. 80143							
$\text{Nb}_5\text{OH-1}(\text{s}) = \text{Nb}_5\text{OH-1}(\text{s}) + \text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	1	KNO3	lgK:-14.6(3SF)	0	MSL	30106
2	25	0	Inf. Dilution	lgK:-12.4(3SF)	1	ELC	

Reaction No. 80144							
$\text{Nb}_5\text{OH-1}(\text{s}) + \text{H}_2\text{O} = \text{Nb}_5\text{OH-1}(\text{s}) + \text{H+1}$							
Note: Temp effect undocumented as far as we know							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0	Inf. Dilution	lgK:-7.74(3SF)	0	CRV	28992
2	19	1	KNO3	lgK:-7.4(2SF)	0	MSL	403
3	19	1	KNO3	lgK:-7.4(2SF)	0	MSL	30106
4	25	0	Inf. Dilution	lgK:-4.9(2SF)	1	ELC	
5	25	0	Inf. Dilution	lgK:-7.39(3SF)	0	CRV	28992

6	25	0	Inf. Dilution	lgK:-6.758 (4SF)	0	CRV	31301
Note (6): > -6.758							
7	25	0.1	NaCl	lgK:-6.322 (4SF)	0	MSL	31301
Note (7): >-6.322							
8	25	0.1	NaCl	lgK:-6.6 (2SF)	0	MSL	31495
9	50	0	Inf. Dilution	lgK:-7.19 (3SF)	0	CRV	28992
10	100	0	Inf. Dilution	lgK:-7.10 (3SF)	0	CRV	28992
11	150	0	Inf. Dilution	lgK:-7.30 (3SF)	0	CRV	28992
12	200	0	Inf. Dilution	lgK:-7.70 (3SF)	0	CRV	28992
13	250	0	Inf. Dilution	lgK:-8.2 (2SF)	0	CRV	28992

Reaction No. 80145							
$\text{Nb+5_OH-1(5)} + \text{H+1} = \text{Nb+5_OH-1(4)} + \text{H2O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:-0.6 (1SF)	0	MSL	403
2	25	0	Inf. Dilution	lgK:1.6 (2SF)	1	ELC	
3	25	0.1	LiClO4	lgK:-3.22 (3SF)	0	MDS	30135

Reaction No. 80146							
$\text{H+1_Na+1(7)_Nb6O19-8_H2O(5)_(s)} = 7\text{<Na+1>} + \text{H+1_Nb6O19-8} + 5\text{<H2O>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.12	NaCl	lgK:-11.64 (4SF)	4	MSL	30112

Reaction No. 80147							
$\text{Nb+5_F-1(5)} + 2\text{<F-1>} = \text{Nb+5_F-1(7)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	5	(H,Na)ClO4	lgK:10.66 (4SF)	0	MSE	5398
2	25	5	NaClO4	lgK:10.7 (3SF)	0	MSE	30113

Reaction No. 80148							
$\text{Nb6O19-8} + 2\text{<H+1>} = \text{H+1(2)_Nb6O19-8}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.97 (0.13SD)	3	MGL	847
Note (1): Application of Bromley theory							
2	25	3	KCl	lgK:23.55 (0.04SD)	3	MGL	847
Note (2): pH from 9.7 to 13.1							

Reaction No. 80149							
Nb6O19-8 + 3<H+1> = H+1(3)_Nb6O19-8							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:39.91(0.18SD)	3	MGL	847
Note (1): Application of Bromley theory							
2	25	3	KCl	lgK:32.90(0.07SD)	3	MGL	847
Note (2): pH from 9.7 to 13.1							

Reaction No. 80150							
Nb+5 + 2<e-1> = Nb+3							
Note: See Nb+5_OH-1(4) + 2<e-1> = Nb+3_OH-1(4) for an attempt at a better representation: the species Nb+5 (and possibly) Nb+3 is considered hopelessly unrealistic							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3.14	H2SO4	E:-0.3730(4SF)	0	MEF	30141
2	25	5.91	H2SO4	E:-0.3849(4SF)	0	MEF	30141
3	25	9.87	H2SO4	E:-0.4261(4SF)	0	MEF	30141

Reaction No. 80151							
Nb+5_OH-1(2) + Tartaric-2 = Nb+5_OH-1(2)_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:25.15(4SF)	2	MGL	906
2	23	0	Unknown	lgK:25.15(2SF)	2	MSL	30357

Reaction No. 80152							
Nb+5_OH-1(2)_Tartaric-2 + Tartaric-2 = Nb+5_OH-1(2)_Tartaric-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Unknown	lgK:7.85(3SF)	2	MGL	906
2	23	0	Unknown	lgK:7.85(2SF)	2	MSL	30357

Reaction No. 80153							
Nb+5_OH-1(2) + 2<Tartaric-2> = Nb+5_OH-1(2)_Tartaric-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	KCl	lgK:33.5(0.7SD)	3	MSL	906
Note (1): From: N.A. Dobvynina et al. Izvest. Akad Nauk SSSR.Khim 1970 2 477							

Reaction No. 80154							
Nb+5_OH-1(2) + 2<Salicylic-2> = Nb+5_OH-1(2)_Salicylic-2(2)							

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.04	Unknown	lgK:22.60(0.04SD)	3	MSP	931
Note (1): NbO3+ notation in [931]							
2	25	0.04	Unknown	lgK:22.6(3SF)	3	MSP	30348
Note (2): NbO3+ notation in the paper							

Reaction No. 80155							
Nb+5_OH-1(4) + H+1(2)_Tartaric-2 = Nb+5_H+1_OH-1(4)_Tartaric-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4.5	(H,Na)Cl	lgK:2.34(3SF)	2	MDS	931
2	20	4.5	(H,Na)Cl	lgK:2.34(3SF)	2	MDS	30350
Note (2): Doubts about how the reaction is written; number of OH in complex in doubt							
3	20	4.5	(H,Na)NO3	lgK:2.33(3SF)	2	MDS	30350
Note (3): Doubts about how the reaction in written; number of OH in complex in doubt							

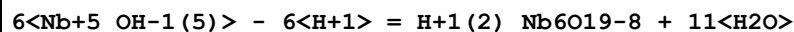
Reaction No. 80156							
Nb+5_OH-1(4) + H+1(2)_Malic-2 = Nb+5_H+1_OH-1(4)_Malic-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4.5	(H,Na)Cl	lgK:2.01(3SF)	2	MDS	931
2	20	4.5	(H,Na)Cl	lgK:2.02(3SF)	2	MDS	30350
Note (2): Doubts about how the reaction is written; number of OH in complex in doubt							
3	20	4.5	(H,Na)NO3	lgK:2.0(2SF)	2	MDS	30350
Note (3): Doubts about how the reaction in written; number of OH in complex in doubt							

Reaction No. 80157							
Nb+5_OH-1(4) + H+1(2)_Succinic-2 = Nb+5_H+1_OH-1(4)_Succinic-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4.5	(H,Na)Cl	lgK:1.53(3SF)	2	MDS	931
2	20	4.5	(H,Na)Cl	lgK:1.48(3SF)	2	MDS	30350
Note (2): Doubts about how the reaction is written; number of OH in complex in doubt							
3	20	4.5	(H,Na)NO3	lgK:1.58(3SF)	2	MDS	30350
Note (3): Doubts about how the reaction is written; number of OH in complex in doubt							

Reaction No. 80162							
6<Nb+5_OH-1(5)> - 5<H+1> = H+1(3)_Nb6O19-8 + 11<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-18.1(3SF)	1	ELC	

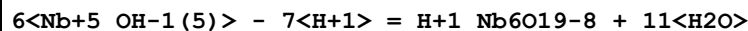
2	25	1	KNO3	lgK:-14.46(4SF)	4	MGL	847
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Reaction No. 80163



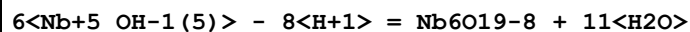
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-29.8(3SF)	1	ELC	
2	25	1	KNO3	lgK:-24.06(4SF)	4	MGL	847

Reaction No. 80164



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-43.0(3SF)	1	ELC	
2	25	1	KNO3	lgK:-33.49(4SF)	4	MGL	847

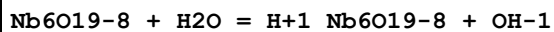
Reaction No. 80165



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KNO3	lgK:-47.04(4SF)	0	MGL	847

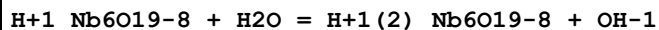
Note (1): Inconsistent with other values in table & with LC

Reaction No. 80166



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	KCl	lgK:11.9(3SF)	0	MGL	30158
2	25	0.1	KCl	lgK:11.98(0.1SD)	0	MGL	30158
3	25	0.5	KCl	lgK:12.17(0.1SD)	0	MGL	30158
4	25	1	KCl	lgK:12.30(0.1SD)	0	MGL	30158
5	25	2	KCl	lgK:12.44(0.1SD)	0	MGL	30158
6	25	3	KCl	lgK:12.60(0.1SD)	0	MGL	30158

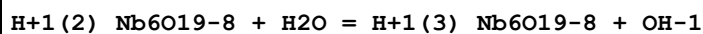
Reaction No. 80167



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	KCl	lgK:11.0(3SF)	0	MGL	30158
2	25	0.1	KCl	lgK:11.01(0.05SD)	0	MGL	30158
3	25	0.5	KCl	lgK:10.97(0.05SD)	0	MGL	30158
4	25	1	KCl	lgK:10.94(0.05SD)	0	MGL	30158

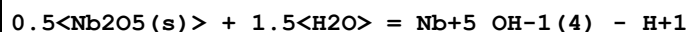
5	25	2	KCl	lgK:10.91(0.05SD)	0	MGL	30158
6	25	3	KCl	lgK:10.86(4SF)	0	MGL	30158

Reaction No. 80168



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	KCl	lgK:10.0(3SF)	0	MGL	30158
2	25	0.1	KCl	lgK:9.94(0.02SD)	0	MGL	30158
3	25	0.5	KCl	lgK:9.81(0.02SD)	0	MGL	30158
4	25	1	KCl	lgK:9.71(0.02SD)	0	MGL	30158
5	25	2	KCl	lgK:9.54(0.02SD)	0	MGL	30158
6	25	3	KCl	lgK:9.39(0.02SD)	0	MGL	30158

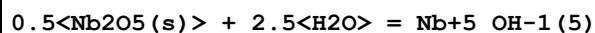
Reaction No. 80170



Note: Solid phase is B-Nb2O5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	1	NaClO4/m	lgK:-7.44(0.06SD)	5	MSL	23700
2	25	0	Inf. Dilution/m	lgK:-7.4(0.2SD)	5	MSL	23700
3	25	0.1	NaClO4/m	lgK:-7.46(0.35SD)	5	MSL	23700
4	25	0.3	NaClO4/m	lgK:-7.64(0.21SD)	5	MSL	23700
5	25	0.5	NaClO4/m	lgK:-7.53(0.14SD)	5	MSL	23700
6	25	1	NaClO4/m	lgK:-7.53(0.09SD)	5	MSL	23700
7	25	6	NaClO4/m	lgK:-8.16(0.05SD)	5	MSL	23700
8	50	1	NaClO4/m	lgK:-7.42(0.11SD)	5	MSL	23700
9	70	1	NaClO4/m	lgK:-7.24(0.18SD)	5	MSL	23700

Reaction No. 80171

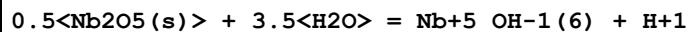


Note: Solid phase is B-Nb2O5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	1	NaClO4/m	lgK:-9.68(0.06SD)	5	MSL	23700
2	25	0	Inf. Dilution/m	lgK:-9.1(0.1SD)	5	MSL	23700
3	25	0.1	NaClO4/m	lgK:-9.20(0.08SD)	5	MSL	23700
4	25	0.3	NaClO4/m	lgK:-9.21(0.07SD)	5	MSL	23700
5	25	0.5	NaClO4/m	lgK:-9.50(0.10SD)	5	MSL	23700

6	25	1	NaClO4/m	lgK:-9.61(0.08SD)	5	MSL	23700
7	25	6	NaClO4/m	lgK:-9.30(0.04SD)	5	MSL	23700
8	50	1	NaClO4/m	lgK:-9.38(0.08SD)	5	MSL	23700
9	70	1	NaClO4/m	lgK:-9.31(0.08SD)	5	MSL	23700

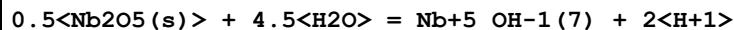
Reaction No. 80172



Note: Solid phase is B-Nb2O5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	1	NaClO4/m	lgK:-16.03(0.11SD)	5	MSL	23700
2	25	0	Inf. Dilution/m	lgK:-14.1(0.3SD)	5	MSL	23700
3	25	0.1	NaClO4/m	lgK:-14.08(0.05SD)	5	MSL	23700
4	25	0.3	NaClO4/m	lgK:-14.39(0.08SD)	5	MSL	23700
5	25	0.5	NaClO4/m	lgK:-14.82(0.11SD)	5	MSL	23700
6	25	1	NaClO4/m	lgK:-15.33(0.11SD)	3	MSL	23700
7	50	1	NaClO4/m	lgK:-15.40(0.21SD)	3	MSL	23700
8	70	1	NaClO4/m	lgK:-14.92(0.12SD)	2	MSL	23700

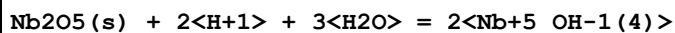
Reaction No. 80173



Note: Solid phase is B-Nb2O5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	1	NaClO4/m	lgK:-24.8(0.2SD)	5	MSL	23700
2	25	0	Inf. Dilution/m	lgK:-23.9(0.6SD)	5	MSL	23700
3	25	0.3	NaClO4/m	lgK:-23.3(0.2SD)	3	MSL	23700
4	25	0.5	NaClO4/m	lgK:-23.9(0.4SD)	3	MSL	23700
5	25	1	NaClO4/m	lgK:-24.6(0.4SD)	3	MSL	23700
6	50	1	NaClO4/m	lgK:-22.8(0.1SD)	1	MSL	23700
7	70	1	NaClO4/m	lgK:-23.0(0.1SD)	5	MSL	23700

Reaction No. 80174

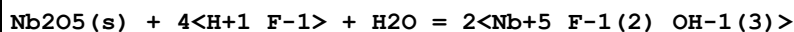


Note: Solid phase is Aldrich Nb2O5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	100	3	NaClO4	lgK:-11.23(0.26SD)	4	MSL	30121
2	200	3	NaClO4	lgK:-10.83(0.37SD)	4	MSL	30121

3	250	3	NaClO4	lgK:-10.86(4SF)	4	MSL	30121
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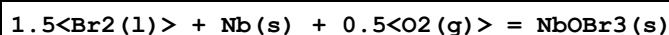
Reaction No. 80175



Note: Solid phase is Aldrich Nb₂O₅(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	150	3	NaClO4	lgK:-3.84(0.20SD)	4	MSL	30121
2	200	3	NaClO4	lgK:-4.04(0.22SD)	4	MSL	30121
3	250	3	NaClO4	lgK:-5.08(0.42SD)	4	MSL	30121

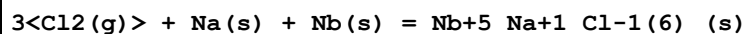
Reaction No. 80176



Note: q

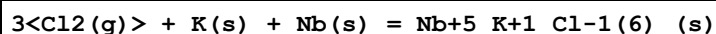
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-179.1(4SF)	4	CRV	22971

Reaction No. 80177



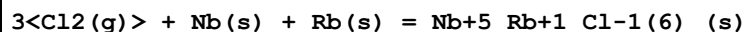
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-291.(3SF)	4	CRV	22971

Reaction No. 80178



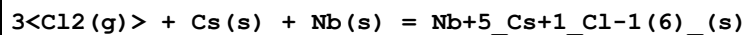
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-301.(3SF)	4	CRV	22971

Reaction No. 80179



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-306.(3SF)	4	CRV	22971

Reaction No. 80180



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-318.(3SF)	4	CRV	22971

Reaction No. 80182

Nb+5_OH-1(5) + H+1(4)_EDTA-4 = Nb+5_OH-1(2)_EDTA-4 + 3<H2O> + H+1							
Note: Impossible to know reaction; I try this one after the one for Ta by the same authors							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1.08	K2SO4	lgK:40.78(4SF)	0	MPL	30356
2	23	0.4	KCl	lgK:39.4(3SF)	0	MPL	30456
Note (2): Rxn. not given in paper							
3	25	0	Inf. Dilution	lgK:40.0(3SF)	1	ESC	30456

Reaction No. 80183							
Nb+5_OH-1(5)_(am.,s) + 3<H+1> + 2<Oxalic-2> = Nb+5_OH-1(2)_Oxalic-2(2) + 3<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.9(3SF)	1	ELC	
2	25	0.5	KNO3	lgK:12.1(3SF)	0	MPH	30355

Reaction No. 80184							
Nb+5_OH-1(5)_(am.,s) + 3<H+1> + 3<Oxalic-2> = Nb+5_OH-1(2)_Oxalic-2(3) + 3<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	KNO3	lgK:17.15(4SF)	2	MPH	30355

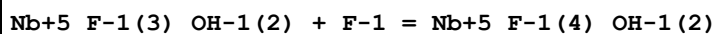
Reaction No. 80185							
Nb+5_OH-1(4) + H+1(2)_Malonic-2 = Nb+5_H+1_OH-1(4)_Malonic-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4.5	(H,Na)Cl	lgK:1.74(3SF)	2	MDS	30350
Note (1): Doubts about how the reaction is written; number of OH in complex in doubt							
2	20	4.5	(H,Na)NO3	lgK:1.71(3SF)	2	MDS	30350
Note (2): Doubts about how the reaction in written; number of OH in complex in doubt							

Reaction No. 80186							
2<Nb6O19-8> + 8<OH-1> + 2<H2O> = 3<OH-1(4)_Nb4O12-4>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.4(1SF)	1	EES	
2	25	1	KCl	lgK:-0.60(1SF)	1	MSP	30136

Reaction No. 80187							
Nb+5_F-1(2)_OH-1(2) + F-1 = Nb+5_F-1(3)_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

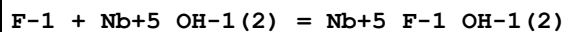
1	25	0.5	NaClO4	lgK:3.80 (3SF)	3	MIS	5398
2	25	0.5	NaClO4	lgK:3.8 (2SF)	3	MIS	30149

Reaction No. 80188



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO4	lgK:4.30 (3SF)	3	MIS	5398
2	25	0.5	NaClO4	lgK:4.3 (2SF)	3	MIS	30149

Reaction No. 80189

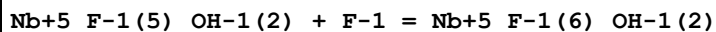


Note: Say Nb+5_OH-1(2), not Nb+5, is what they had; see F-1 + Nb+5 = Nb+5_F-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	17	H2SO4	lgK:7.12 (3SF)	0	MSP	5398

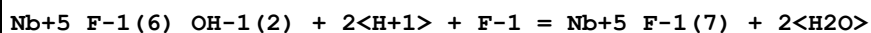
Note (1): Nb+5 unreal species

Reaction No. 80190



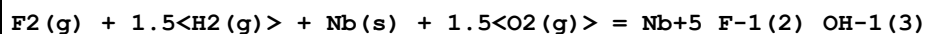
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO4	lgK:4.67 (3SF)	3	MIS	5398
2	25	0.5	NaClO4	lgK:4.7 (2SF)	3	MIS	30149

Reaction No. 80191



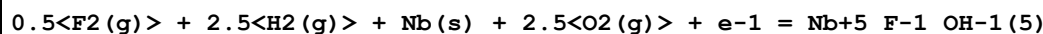
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO4	lgK:11.4 (3SF)	3	MIS	30149

Reaction No. 80197



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-377.770 (3SF)	3	MDG	30153
2	25	0	Inf. Dilution	dH:-407.960 (3SF)	3	MDG	30153

Reaction No. 80198



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-403.480 (3SF)	3	MDG	30153

2	25	0	Inf. Dilution	dH:-453.910 (3SF)	3	MDG	30153
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Reaction No. 80199							
$K(s) + Nb(s) + 1.5O_2(g) = KNbO_3(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1261.5 (10.0SD) kJ	3	EOR	30153
Note (1): Original Russian reference given							
2	25	0	Inf. Dilution	dH:-1346.8 (5SF) kJ	3	EOR	30153
Note (2): Original Russian reference given							

Reaction No. 80200							
$Na(s) + Nb(s) + 1.5O_2(g) = NaNbO_3(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1207.6 (6.3SD) kJ	3	EOR	30153
Note (1): Original Russian reference given							
2	25	0	Inf. Dilution	dH:-1293.7 (5.4SD) kJ	3	EOR	30153
Note (2): Original Russian reference given							

Reaction No. 80201							
$Li(s) + Nb(s) + 1.5O_2(g) = LiNbO_3(s)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1279.1 (5SF) kJ	3	EOR	30153
Note (1): Original Russian reference given							
2	25	0	Inf. Dilution	dH:-1364.9 (5SF) kJ	3	EOR	30153
Note (2): Original Russian reference given							

Reaction No. 80202							
$Nb+5_{OH-1}(4) + 2e-1 = Nb+3_{OH-1}(4)$							
Note: Both Nb+5 and Nb+3 species assumed so that the E0 from $Nb+5 + 2e-1 = Nb+3$ can be better represented (just hopefully)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.7 (2SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.39 (2SF)	0	EES	

Reaction No. 80203							
$Nb+3 + 4OH-1 = Nb+3_{OH-1}(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:34.(2SF)	1	EES	
Note (1): By analogy with Al+3							

Reaction No. 80207							
Nb+3 + 2<OH-1> = Nb+3_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.(2SF)	1	EES	
Note (1): By analogy with Al+3_OH-1(2)							

Reaction No. 80208							
Nb+5_F-1(2)_OH-1(2) + 2<OH-1> = Nb+5_OH-1(4) + 2<F-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.7(3SF)	1	ELC	

Reaction No. 80210							
Nb+5_OH-1(2)_Oxalic-2(3) = Nb+5_OH-1(2)_Oxalic-2(2) + Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Self Medium	lgK:6.17(0.04SD)	2	MSL	30164
Note (1): pH 0.3-1.8							

Reaction No. 80215							
Nb+5 + OH-1 = Nb+5_OH-1							
Note: Nb+5 is considered too highly charged to exist							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:33.2(3SF)	0	EQC	30106
Note (1): Highly dubious calculation method							
2	25	1	KNO3	lgK:33.2(3SF)	0	EES	

Reaction No. 80216							
Nb+5_OH-1 + OH-1 = Nb+5_OH-1(2)							
Note: Nb+5_OH-1 is considered too highly charged to exist							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:29.8(3SF)	0	EQC	30106
Note (1): Highly dubious calculation method							
2	25	1	KNO3	lgK:29.8(3SF)	0	EES	

Reaction No. 80217							
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Nb+5_OH-1(2) + OH-1 = Nb+5_OH-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:25.2(3SF)	0	EQC	30106
Note (1): Highly dubious calculation method							
2	25	0	Inf. Dilution	lgK:24.5(3SF)	1	EDH	
3	25	1	KNO3	lgK:23.5(3SF)	1	EES	

Reaction No. 80218							
Nb+5_OH-1(3) + OH-1 = Nb+5_OH-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	19	1	KNO3	lgK:19.9(3SF)	2	EQC	30106
Note (1): Highly dubious calculation method							
2	25	1	KNO3	lgK:19.9(3SF)	1	EES	

Reaction No. 80698							
H+1(2)_Nb6O19-8 + H+1 = H+1(3)_Nb6O19-8							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.63(4SF)	4	CRV	31495

Reaction No. 80801							
4<Nb+5_OH-1(8)> = Nb4O16-12 + 16<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.3(2SF)	1	EES	
2	25	1	KCl	lgK:8.8(2SF)	1	MSP	30136
Note (2): Nb+5_OH-1(8) written as NbO2_OH-1(4) in publication							

Reaction No. 80802							
OH-1(4)_Nb4O12-4 + 4<OH-1> + 12<H2O> = 4<Nb+5_OH-1(8)>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-16.5(3SF)	1	ELC	
2	25	1	KCl	lgK:-11.96(4SF)	0	MSP	30136
Note (2): Nb+5_OH-1(8) written as NbO2_OH-1(4) in publication							

Reaction No. 80803							
0.5<H+1(2)_Nb12O36-12> + 3<OH-1> = Nb6O19-8 + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.0(3SF)	1	EES	

2	25	1	KCl	lgK:17.92 (2SF)	2	MPT	30156
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Reaction No. 80804							
$H+1(2)_{Nb12O36-12} + 26<OH-1> = 3<Nb4O16-12> + 14<H2O>$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16. (2SF)	1	EES	
2	25	1	KCl	lgK:15.24 (2SF)	2	MPT	30156

Reaction No. 80805							
$H+1(2)_{Nb12O36-12} + 74<OH-1> = 12<NbO6-7> + 38<H2O>$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:429.60 (2SF)	2	MPT	30156

Reaction No. 80806							
$2<H+1(3)_{Nb6O19-8}> = H+1(2)_{Nb12O36-12} + 2<H2O>$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-33.8 (3SF)	1	ELC	
2	25	1	KCl	lgK:23.46 (4SF)	0	MPT	30156

Reaction No. 80807							
$2<H+1(3)_{Nb6O19-8}> + 5<H+1> = H+1(7)_{Nb12O36-12} + 2<H2O>$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:61.03 (2SF)	2	MPT	30156

Reaction No. 80808							
$2<H+1(3)_{Nb6O19-8}> + 7<H+1> = H+1(9)_{Nb12O36-12} + 2<H2O>$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:66.84 (2SF)	2	MPT	30156

Reaction No. 80809							
$2<H+1(3)_{Nb6O19-8}> + 8<H+1> = H+1(10)_{Nb12O36-12} + 2<H2O>$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	KCl	lgK:78.03 (2SF)	2	MPT	30156

Reaction No. 80810							
$Nb+5_{OH-1(5)}(am.,s) + H2O = Nb+5_{OH-1(6)} + H+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-12.5 (3SF)	1	ELC	

2	25	0	Inf. Dilution	lgK:-14.457(5SF)	0	MSL	31301
Note (2): value/2; Orig: Nb2O5(pr)+7H2O=SNB(OH)6--+2H+, here amorphous Nb+5_OH-1(5)_s)							
3	25	0.1	NaCl	lgK:-14.243(5SF)	0	MSL	31301
Note (3): value/2; Orig: Nb2O5(pr)+7H2O=SNB(OH)6--+2H+, here amorphous Nb+5_OH-1(5)_s)							

Reaction No. 81030							
$3\text{H}_2(\text{g}) + \text{Nb}(\text{s}) + 3\text{O}_2(\text{g}) + \text{e}^- = \text{Nb}+5_ \text{OH}-1(6)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:286.2(4SF)	1	ELC	
2	25	1	Unknown	dG:-1448.3(0.100SD)kJ	4	CRV	32222

Reaction No. 81031							
$\text{NbO}_3-1 + \text{H}_2\text{O} + 2\text{H}^+ = \text{Nb}+5_ \text{OH}-1(4)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.5(2SF)	1	ELC	
2	25	1	Unknown	lgK:6.90(0.03SD)	0	CRV	32222

Reaction No. 81032							
$\text{NbO}_3-1 + 2\text{H}_2\text{O} + \text{H}^+ = \text{Nb}+5_ \text{OH}-1(5)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.13(2SF)	1	ELC	
2	25	1	Unknown	lgK:7.34(0.03SD)	0	CRV	32222

Reaction No. 81033							
$2\text{NbO}_3-1 + 2\text{H}^+ = \text{Nb}_2\text{O}_5(\text{s}) + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:24.34(0.04SD)	0	CRV	32222

Reaction No. 81034							
$\text{NbO}_3-1 + 2\text{H}^+ + \text{e}^- = \text{NbO}_2(\text{s}) + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.7(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:7.98(0.03SD)	0	CRV	32222

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
ELC	Linear Combination of Reactions
ENS	Not specified
EOD	Calculated from data of others
EOR	Other Reaction Constants
EQC	Quintuply charged central ion method
ESC	Standard conditions anchor
MCL	Calorimetry
MDG	Combination of Thermodynamic Data
MDS	Distribution between two phases
MEF	Electromotoric force, not specified
MGL	Glass electrode
MIS	Ion selective electrode
MPH	pH method, not specified

MPL	Polarography
MPT	Potentiometry
MSE	Solvent Extraction
MSL	Solubility
MSP	Spectroscopy
MTD	Temperature difference

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